Spoken Language Identification for 22 Indian Languages: Fine-Tuning MMS-300m with Audio Augmentation and Adversarial Debiasing

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Abstract

Spoken language identification (LID) across India's 22 constitutionally recognized languages is a challenging task due to acoustic similarities between related languages and severe scarcity of training data. In this work, we fine-tune the Massively Multilingual Speech model MMS-300m (Meta AI) on a constrained dataset of 400 samples per language from only 5 speakers per language. The limited speaker diversity introduces a critical speaker bias risk, where the model may learn speaker identity rather than linguistic features. To mitigate this, we develop an audio augmentation pipeline comprising pitch shifting and speed perturbation, and additionally implement Domain Adversarial Neural Networks (DANN) (Ganin and Lempitsky, 2014) as a model-based debiasing approach. Our augmented model achieves 37.58% top-1 accuracy, and combining augmentation with DANN achieves **41.09%**, surpassing the 33.39% baseline by 7.7 percentage points. Embedding analysis via t-SNE and linear probing reveals that augmentation and DANN are complementary: augmentation most directly reduces speaker-encodable information, while DANN produces richer language-discriminative representations.

1 Introduction

India is known for its diversity in linguistics all around the world, with 22 languages officially recognized by the consititution which span multiple families of languages(Wikipedia contributors). Building robust spoken language identification (LID) systems for such a diverse set is critical for real-world applications.

The task poses two interrelated challenges. First, many Indian languages share phonetic and prosodic characteristics, making acoustic discrimination non-trivial. Second, the available training data is severely limited: only 400 samples and 5 speakers per language (badrex). With so few speakers, a model risks overfitting to speaker-specific vocal characteristics rather than learning language-discriminative features, a problem we refer to as *speaker bias*.

Pre-trained self-supervised speech models such as MMS (Meta AI) have demonstrated strong transfer learning for downstream speech tasks including LID. In this work, we address both challenges. We fine-tune MMS-300m (Meta AI). To combat speaker bias, we design an audio augmentation pipeline. Audio augmentation, particularly pitch shifting and speed perturbation is a standard approach to artificially expand speaker variability (PyTorch/torchaudio). We further implement Domain Adversarial Neural Networks (DANN) (Ganin and Lempitsky, 2014) to explicitly suppress speaker identity in the learned representations via a gradient reversal layer, a principle also applied to remove demographic attributes from text representations (Elazar and Goldberg, 2018).

2 Methodology

2.1 Dataset and Baseline Setup (Task 1)

We utilize the NNTI Indian speech dataset (badrex), featuring 22 languages across four families: Indo-Aryan, Dravidian, Sino-Tibetan, and Austroasiatic. The data is extremely sparse: each language has 400 samples from only 5 speakers (110 total speakers). This scarcity introduces significant *speaker bias*, where models exploit speaker-specific shortcuts (pitch, timbre) rather than linguistic features.

After evaluating candidates such as `mHuBERT-147` and `w2v-BERT-2.0`, we selected `facebook/mms-300m` (Meta AI) as our backbone due to its strong performance and computational efficiency. For Task 1, our initial baseline achieved 31.90% accuracy. By further tuning the hyperparameters, we established an *extended baseline* that reached 33.39% top-1 accuracy. This tuned configuration is utilized as the shared foundation for all subsequent downstream tasks. Table 1 contrasts the hyperparameters of the initial baseline with the extended baseline.

Hyperparameter	Initial Baseline	Extended Baseline
Tuned Parameters		
Learning rate	3×10^{-5}	5×10^{-5} (Cosine decay)
Weight decay	0.0	0.01
Max epochs	8	15
Early stopping	None	Yes (patience=7)
Dropout	None	0.1
Constant Parameters		
Model	facebook/mms-300m	
Optimizer	AdamW	
Effective batch size	32 (8×4 grad. accum.)	
Precision	bf16	
Audio max length	7 seconds	
Warm-up ratio	0.08	

Table 1: Hyperparameter configurations for the initial baseline and the tuned extended baseline applied to all downstream tasks.

2.2 Debiasing and Adversarial Training (Task 2)

To mitigate speaker bias, we implemented two complementary strategies:

Audio Augmentation: A data-centric pipeline where 70% of samples undergo random pitch shifting ($n \sim \mathcal{U}(-4, 4)$ semitones) and speed perturbation ($f \sim \mathcal{U}(0.85, 1.15)$) (PyTorch/torchaudio). This artificially increases speaker diversity while preserving linguistic content.

We used a special kind of neural network called Domain Adversarial Neural Network (DANN) (Elazar and Goldberg, 2018) with the MMS-300m encoder. This network takes the last 4 layers of the transformer and combines them to make a 1024-dimensional embedding. Then, it sends this embedding to two different parts: a language classifier that uses a simple linear model, a speaker adversary consisting of a 2-layer MLP (1024→256→110) with ReLU activation and a dropout of 0.1. The speaker adversary helps the network learn to ignore the speaker's voice and focus on the language.

A Gradient Reversal Layer (GRL) inserted between the encoder and speaker head negates the gradient from the adversary by a factor λ, training the encoder to be language-discriminative but speaker-uninformative. The adaptation factor is annealed as $\lambda(p) = \frac{2}{1+e^{-10p}} - 1$, where p is the training progress. The final objective is $\mathcal{L} = \mathcal{L}_{\text{lang}} + \mathcal{L}_{\text{spk}}$.

3 Experiments and Results

3.1 Main Results (Tasks 1 & 2)

Table 2 compares model validation performance of the three models, namely, ExtendedBaseline, ExtendedBaseline+augmentation and ExtendedBaseline+DANN+augmentation. As per Task 1 instruction, the `MMS-300m` model was trained to official `MMS-300m` baseline reported 31.96% accuracy without . Our tuned baseline(as per Table 1) reached 33.39% (Macro F1: 0.321). While exceeding chance (4.54%), this signifies the difficulty of separating languages that are similar (with sparse data).

For the sake of clarity in the following manuscript, ExtendedBaseline will be referred to as Baseline as we perform all our downstream task on it.

Considering audio augmentation alone provided improvement of accuracy to 37.58% (Macro F1: 0.370),which proves that expanding acoustic diversity successfully limits speaker memorization. Combination of DANN with augmentation resulted in the best overall performance at **41.09%** (Macro F1: 0.400). This 7.7 percentage point gain compared to the baseline demonstrates improvement of joint data and model-centric debiasing.

Note that DANN's combined training loss appears artificially high because the adversarial objective inherently adds speaker classification loss. Consequently, raw loss values are not directly comparable across models, making validation accuracy the primary evaluation metric.

System	Accuracy	Macro F1
Chance (random)	4.54%	–
Baseline (no aug, no DANN)	33.39%	0.321
Augmentation only	37.58%	0.370
DANN + Augmentation	**41.09%**	**0.400**

Table 2: Validation set results. All fine-tuned models use identical hyperparameters; differences reflect the debiasing strategy only.

3.2 Per-Language Analysis

Our combination of DANN and augmentation setup improves the F1 scores for most languages (Table 3). Distinct languages like Bodo (+0.199) and Assamese (+0.098) saw the biggest gains. However, performance dropped for Hindi and Maithili. Because these languages share so much phonology with Urdu, stripping away speaker traits seems to also wipe out the subtle acoustic clues needed to tell them apart.

Language	Base	Aug	DANN+Aug
Kashmiri	0.753	0.713	**0.746**
Manipuri	0.592	0.565	**0.647**
Bodo	0.427	0.559	**0.626**
Assamese	0.538	0.571	**0.636**
Malayalam	0.525	0.576	**0.586**
Sanskrit	0.239	**0.627**	0.615
Nepali	0.400	0.279	**0.511**
Urdu	0.347	0.413	**0.504**
Maithili	0.237	0.205	0.148
Hindi	0.083	0.145	0.069
Bengali	0.087	0.158	**0.227**

Table 3: Per-language F1 scores (selected). Bold = best per language.

3.3 Confusion Matrix Analysis

Examining model errors (Fig 1), the baseline reveals heavy confusion within the Dravidian cluster (Tamil as Telugu 52 times). DANN+Aug (Task 2) reduces this to 35. However, DANN worsens Indo-Aryan overlaps; for instance, Hindi samples are absorbed by Urdu and Punjabi, with only 4/150 Hindi samples correctly identified. Despite these linguistic similarities causing issues, DANN significantly improves accuracy for distinct languages like Kashmiri and Manipuri.

4 Embedding Analysis (Task 3)

4.1 Method

We extract utterance-level embeddings from the final encoder layer via a forward hook and mean-pool over time, yielding a 3300×1024 matrix from the validation set. We apply:

1. **t-SNE** (perplexity=30, 1000 iterations) coloured by language and speaker identity.

2. **Linear probe**: 5-fold cross-validated logistic regression to predict speaker identity (110 classes) and language (22 classes). Lower speaker probe indicates less speaker-encoded information; higher language probe indicates more language- discriminative representations.

4.2 Probe Results

Augmentation alone achieves the lowest speaker probe (0.485), indicating that acoustic perturbation effectively disrupts speaker-discriminative cues in

Model	Accuracy	Speaker Probe	Lang. Probe
Baseline	33.39%	0.511 ± 0.016	0.621 ± 0.013
Aug only	37.58%	$\mathbf{0.485} \pm 0.016$	0.640 ± 0.015
DANN+Aug	**41.09%**	0.521 ± 0.017	$\mathbf{0.646} \pm 0.011$

Table 4: Linear probe accuracy (5-fold CV). Lower speaker probe = less speaker bias. Higher language probe = more language-discriminative representations.

the embeddings . DANN+Aug shows a slightly higher speaker probe (0.521).

DANN's adversarial training prevents the *language head* from exploiting speaker shortcuts at training time, but does not directly minimise speaker information in the shared encoder. The encoder under DANN learns richer representations,evidenced by the highest language probe (0.646),which incidentally contain more speaker-predictable information. Augmentation directly destroys acoustic variation at the input, reducing speaker information in all downstream representations.

The two methods are complementary: augmentation reduces speaker-encodable information; DANN redirects learning toward language-discriminative features. Their combination achieves the best accuracy.

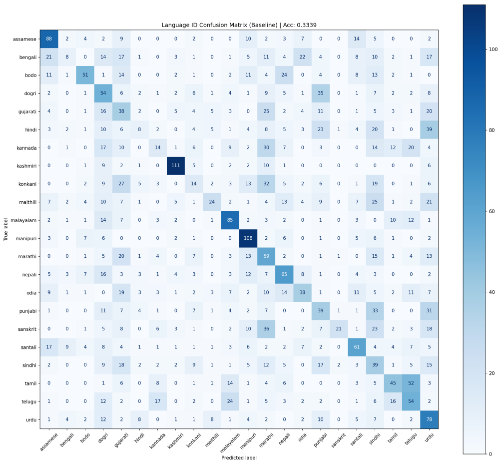
4.3 t-SNE Analysis

The baseline t-SNE (Figure 2a) shows only partial language structure: Kashmiri forms a weak cluster while most Indo-Aryan languages form an undifferentiated mass. The augmentation model (b) shows more visible groupings. DANN+Aug (c) achieves the clearest separation, with Kashmiri, Manipuri, and Assamese forming compact, well-separated clusters.

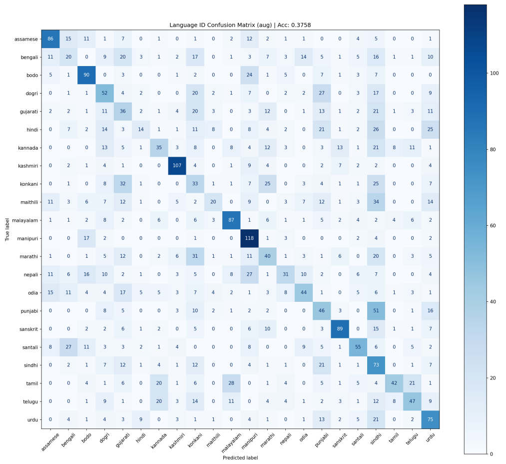
All three speaker t-SNE plots show largely scattered distributions without strong per-speaker clusters (Figure 3), indicating that none of the models form globally coherent speaker-identity manifolds. The linear probe provides a more sensitive quantitative measure of residual speaker encoding than visual inspection alone.

5 Conclusion

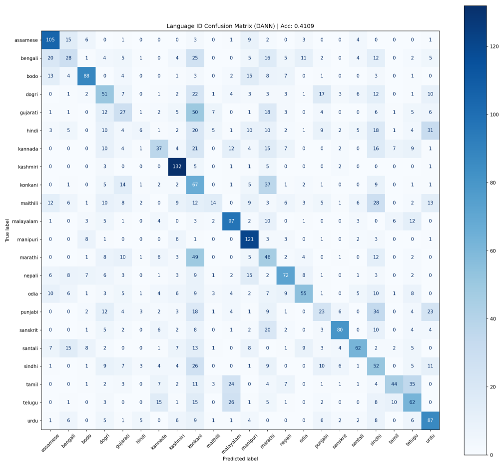
We presented a spoken language identification system for 22 Indian languages by fine-tuning MMS-300m with audio augmentation and adversarial speaker debiasing via DANN. Augmentation alone achieves 37.58% top-1 accuracy; combining augmentation with DANN achieves **41.09%**, a 7.7



(a) Baseline (33.39%)



(b) Aug only (37.58%)



(c) DANN+Aug (41.09%)

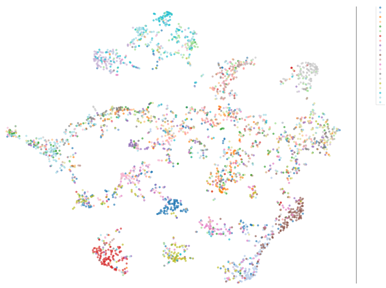
Figure 1: Confusion matrices across all three configurations. DANN strengthens diagonal entries for acoustically distinctive languages (Kashmiri, Manipuri, Assamese) but degrades phonologically similar Indo-Aryan pairs (Hindi, Maithili).

percentage point gain over the 33.39% baseline. Embedding analysis shows that augmentation and DANN are complementary: augmentation most directly reduces speaker-encodable information in representations, while DANN produces richer language-discriminative features as measured by linear probing. Confusion analysis shows that remaining errors concentrate along linguistic family boundaries, particularly within the phonologically similar Indo-Aryan cluster. Future work should explore more diverse speaker data and larger pretrained models to further improve language discrimination.

Limitations

While our debiasing strategies improve overall performance, several limitations remain. First, with only 5 speakers per language, stochastic augmentation cannot fully substitute for genuine speaker diversity in the training data. Second, our exploration was constrained by the 600M parameter limit; we could not evaluate whether larger architectures (e.g., w2v-BERT-2.0) naturally suppress speaker bias better than MMS-300m without explicit adversarial branches. Linguistically, our debiasing approach struggles to separate near-identical language pairs (such as Hindi and Urdu), as suppressing acoustic speaker cues inadvertently removes the subtle phonetic signals required to distinguish them. Finally, results are reported on a single validation split; cross-validation over multiple speaker splits would provide more robust generalization estimates.

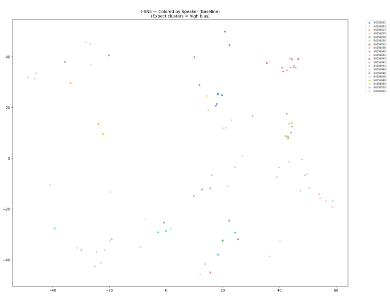
(a) Baseline



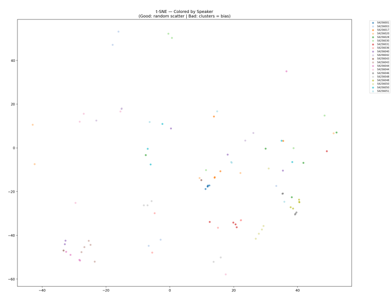
(b) Aug only

(c) DANN+Aug

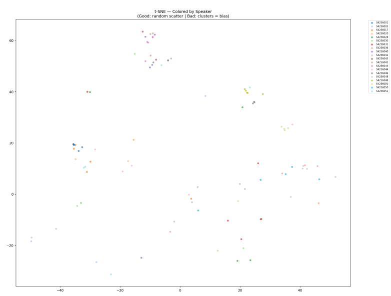
Figure 2: t-SNE coloured by language. DANN+Aug produces the tightest clusters for distinctive languages (Kashmiri, Manipuri, Assamese) while the Indo-Aryan core remains mixed across all models.



(a) Baseline



(b) Aug only



(c) DANN+Aug

Figure 3: t-SNE coloured by speaker identity (20 of 110 shown). All three models show scattered distributions with no strong per-speaker clusters, indicating limited global speaker structure in the embedding space.

Ethics Statement

This work uses a publicly available speech dataset for academic research only. The system performs language identification and does not process personally identifiable information. Care should be taken when deploying such systems in real-world settings, as language misclassification could negatively affect speakers of minority languages, particularly those already underrepresented in speech technology.

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